JIAJIE JIAN

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EDUCATION

◦ **Chongqing University** Chongqing, China

M.Eng. in Information and Communication Engineering Sep. 2024 – Jun. 2027 (expected)

- **Co-advisors:** Prof. Shu Fu & Prof. Min Liu
- **Average grade:** 88.57/100
- **Selected Coursework:** Modern Signal Processing, Information Theory and Coding, Deep Learning, Radar Systems and Signal Processing

◦ **Sichuan Normal University** Chengdu, China

B.Eng. in Electronic Information Engineering Sep. 2020 – Jun. 2024

- **Thesis Advisor:** Dr. Jialin Ding
- **Overall average:** 84.16/100
- **Selected Coursework:** Signals and Systems, Digital Signal Processing, Principles of Communications, Probability and Statistics, Linear Algebra

RESEARCH INTERESTS

Integrated Sensing and Communications (ISAC); Physical-Layer Security; Optimization & Machine Learning for Wireless Communications; Information Theory; Statistical Signal Processing.

PUBLICATIONS AND MANUSCRIPTS

Journal Articles

[J1] **Jiajie Jian**, Min Liu, Shu Fu, and Yongming Huang, "Bidirectional Covertness and Security Enhancement in FD-ISAC Systems," *IEEE Transactions on Vehicular Technology*, 2026. [DOI]

Manuscripts

[M1] **Jiajie Jian**, Zan Li, Shu Fu, Jiangbo Si, and Min Liu, "Cross-Boundary Covert UAV Service Near No-Fly Zones: Risk-Aware Trajectory and Scheduling," *IEEE Transactions on Mobile Computing*, under review.

[M2] **Jiajie Jian**, Shu Fu, Tingting Chen, and Min Liu, "Resilient Beamforming for Anti-Jamming MIMO-ISAC," *IEEE Wireless Communications Letters*, submitted.

RESEARCH EXPERIENCE

◦ **Sensing-Guided Beamforming for MIMO-ISAC under Unknown Jamming [M2]** Feb. 2026 – Present

- Introduced a sensing-guided beamforming method that infers a surrogate spatial-interference model from sensing observations, mitigating communication and tracking losses without prior knowledge of the jammer's strategy.
- Reduced the beamforming update to a two-variable dual search by exploiting problem's Kronecker-product structure, cutting the number of optimization variables by more than **95%** and runtime by more than **86%** relative to a direct CVX implementation.
- Built a lightweight learning model that adaptively tunes anti-jamming design parameters from sensing feedback, improving effective sum rate by **18%** over fixed robust designs under reactive jamming.
- Compared the method with representative baselines and conducted ablation studies across jamming intensities, showing that null broadening and dedicated sensing degrees of freedom were the main contributors to robustness.

- **Diffusion-Assisted EKF Tracking for UAV-ISAC Beam Alignment** Mar. 2026 – Jun. 2026
  - Augmented an EKF tracker with a diffusion model for agile UAV maneuvers, using historical states to forecast nonlinear motion and reduce tracking drift.
  - Adapted the DDIM denoising depth online using the normalized innovation squared (NIS), suppressing tracking drift under model mismatch while avoiding unnecessary denoising steps.
  - Implemented an end-to-end PyTorch pipeline spanning model training, DDIM inference, and EKF integration, and contributed the resulting state-estimation module to a collaborative UAV-ISAC beamforming study.
  - Validated the method on 50 deterministic UAV trajectories, reducing position and angular RMSE by at least **54%** in forecasting and approximately **20–27%** in continuous tracking relative to EKF; these gains improved multi-step beam alignment, increasing normalized beamforming gain from **0.892** to **0.991** and reducing 1-dB outage from **13.7%** to **0.8%**.
- **Risk-Aware UAV Trajectory and Scheduling for Wireless Service [M1]** Sep. 2025 – Present
  - Integrated UAV trajectory planning with user scheduling to maintain reliable wireless service near regulated areas with limited freedom to maneuver, while controlling RF leakage throughout the mission.
  - Developed a mission-level RF-leakage risk model that aggregates slot-level leakage events into an additive risk budget expressed as a function of UAV trajectory, user scheduling, and transmission-rate decisions.
  - Derived closed-form per-slot scheduling updates from the budget's additive structure via Lagrangian relaxation, then jointly updated the trajectory, rate, and leakage variables using successive convex approximation.
  - Evaluated the joint design through key-parameter sweeps and 2-D visualizations of optimized UAV trajectories and user scheduling, achieving $>$**40%**total spectral efficiency gains over heuristic scheduling and about **15%** over an optimized fixed-trajectory baseline.
- **Joint Transceiver Design for Secure Full-Duplex ISAC [J1]** Dec. 2024 – Aug. 2025
  - Devised a joint FD-ISAC transceiver design for bidirectional security, limiting downlink exposure from radar sensing while leveraging public uplink traffic as cover for confidential uplink transmission, and controlling interference at the legitimate receiver.
  - Applied detection-metric monotonicity to separate surveillance-channel uncertainty from beamforming and power allocation; then converted the robust constraints into linear matrix inequalities via the S-procedure and approximated remaining nonconvex terms with tight convex surrogates.
  - Derived a zero-forcing implementation that separates beam directions from power allocation, reducing computational complexity **from** $\mathcal{O}(N^{6.5})$for matrix optimization **to** $\mathcal{O}(N^2)$ for a two-variable power search.
  - Conducted simulations across security, sensing, and channel-uncertainty regimes, achieving bidirectional protection through native-signal reuse and embedded artificial noise, with no dedicated security hardware.
  - Developed a follow-on **information-theoretic** analysis of the FD-ISAC framework, deriving an achievable covert throughput coefficient and quantifying the downlink masking–self-interference tradeoff. [proof sketch]

SKILLS

- **Methods:** Optimization Theory, Statistical Signal Processing, Machine Learning, Matrix Analysis
- **Programming & Tools:** Python (PyTorch), MATLAB (CVX), LaTeX
- **Languages:** English (IELTS Academic: 7.5); Mandarin Chinese (native)

HONORS & AWARDS

- First-Class Academic Scholarship, Chongqing University 2024, 2025
- First and Second-Class Academic Scholarship, Sichuan Normal University 2021, 2023